SwitchMixBench: Evaluating Multilingual Language Model Robustness under Code-Switching Noise

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Abstract

Multilingual foundation models are typically evaluated on clean, monolingual benchmarks that underestimate the distributional shifts present in real-world multilingual usage. In many deployment settings, user inputs contain code-switching and informal noise that may affect model performance despite preserving sentence-level semantics.

We introduce SwitchMixBench, an open-source benchmark for evaluating the robustness of multilingual models under controlled code-switching perturbations. SwitchMixBench constructs paired clean and perturbed inputs designed to preserve task labels while introducing multilingual lexical mixing and lightweight informal noise. This enables direct estimation of robustness gaps under label-preserving distribution shift.

Empirically, we observe substantial degradation in task accuracy on Natural Language Inference when moving from clean to code-switched inputs. To investigate the underlying mechanism, we complement performance evaluation with tokenizer-level and representation-level analysis. Code-switching induces measurable changes in subword segmentation and token distributions, resulting in sequence inflation and altered token boundaries. However, layer-wise analysis reveals that global sentence-level contextual representations remain nearly invariant across encoder layers.

This decoupling between tokenizer-level perturbation and latent representation stability suggests that robustness failures arise downstream of contextual encoding. We hypothesize that localized token-level perturbations may disrupt attention alignment or task-specific decision boundaries without inducing large-scale embedding drift.

Taken together, these findings highlight the importance of evaluating multilingual models under realistic code-switching conditions and suggest that robustness failures may emerge independently of global representation instability.

1 Introduction

Multilingual foundation models such as BERT [3] and XLM-R [2] have become standard components in modern NLP systems. They are used in search, chatbots, translation-assisted workflows, education tools, and information access pipelines across a wide variety of languages. In practice, however, deployed systems rarely receive "clean" monolingual inputs. Instead, real user text frequently contains phenomena such as informal spelling, missing diacritics, creative abbreviations, and, crucially, *code-switching*—the alternation between languages within the same message or even within the same clause.

Code-switching is not a niche edge case. It is a natural communication strategy in multilingual communities, where speakers adapt dynamically to context, audience, and lexical availability. In West Africa, for example, daily communication often mixes French with local languages, and this multilingual reality is underrepresented in standard evaluation suites. This creates an *evaluation mismatch*: models are typically assessed on carefully curated datasets that underestimate the distributional shifts present in deployment.

Problem. Current multilingual evaluation protocols largely assume that inputs are grammatical, monolingual, and orthographically standardized. As a result, a model can appear strong on benchmarks while failing

on real user traffic that contains light multilingual mixing. This mismatch becomes particularly problematic when such systems are used for high-stakes access to information (education, medical or administrative guidance, etc.), where reliability under realistic text is critical.

Goal. This project aims to quantify and analyze robustness failures of multilingual models under controlled code-switching and informal noise, while maintaining semantic equivalence. The central idea is to construct *paired* evaluation inputs: for each clean example x, we generate a perturbed counterpart $\tilde{x}$ such that the gold label remains unchanged. This enables direct measurement of robustness gaps without confounding factors from label noise or dataset drift.

SwitchMixBench. We introduce SwitchMixBench, an open-source benchmark and evaluation toolkit designed for reproducible robustness measurement under code-switching. SwitchMixBench supports multiple tasks (currently Natural Language Inference and Question Answering) and provides: (i) a controlled perturbation pipeline, (ii) a standardized evaluation interface, and (iii) analysis modules that move beyond "accuracy drops" by probing the *mechanisms* by which perturbations propagate through the system.

Beyond descriptive evaluation. A key objective of this work is to go beyond reporting that performance decreases. We complement performance evaluation with:

- **Tokenizer-level distribution shift:** how perturbations alter subword segmentation and token distributions;
- **Latent representation analysis:** whether global sentence representations drift under perturbations;
- **Efficiency implications:** whether robustness failures are accompanied by computational overhead (sequence inflation, activation memory estimates, latency).

Main findings (high-level). Empirically, we observe a substantial robustness gap on NLI when moving from clean to SwitchMix inputs. Mechanistically, we observe measurable tokenizer-level shifts, while sentence-level latent representations remain nearly unchanged. This decoupling suggests that robustness failures arise downstream of global encoding, likely via localized token-level interactions that influence decision boundaries.

Reproducibility. SwitchMixBench is released publicly with configuration-driven scripts that generate datasets, run evaluations, and export analysis tables. The code is designed to support extension to additional language pairs and perturbation families.

Paper organization. Section 2 summarizes the contributions of SwitchMixBench. Section 3 situates the benchmark within prior work on multilingual robustness and input perturbation. Section 4 introduces the paired evaluation framework and the SwitchMix perturbation pipeline. Section 5 describes the experimental setup, evaluation protocol, and diagnostic analyses. Section 6 reports task performance and mechanistic findings at the tokenizer, representation, and efficiency levels. Sections 7–10 discuss methodological limitations, deployment implications across application domains, and an interpretation of robustness failures under multilingual perturbation.

2 Contributions

Our contributions are:

- **Benchmark for controlled code-switching robustness.** We introduce SwitchMixBench, a benchmark for evaluating multilingual model robustness under controlled code-switching and informal noise perturbations, constructed as paired clean/perturbed inputs with label preservation.
- **Config-driven generation and evaluation pipeline.** We provide a reproducible pipeline that (i) builds datasets into standardized JSONL formats, (ii) runs task evaluations under identical inference settings, and (iii) exports result tables for analysis.

- **Mechanistic analysis beyond performance drops.** We augment benchmark evaluation with diagnostic analysis modules that quantify: (i) tokenizer-level distribution shift, (ii) latent representation stability across encoder layers, and (iii) efficiency implications (sequence length inflation, activation memory estimates, latency).
- **Open-source release for research and extension.** We release code, configurations, and documentation to enable replication and extension to additional language pairs, perturbation types, and tasks.

What this paper is (and is not). SwitchMixBench is designed as a controlled diagnostic instrument. It is not a claim that existing models cannot process multilingual content in general, but rather a targeted evaluation framework that reveals brittle behavior under realistic multilingual mixing, and provides measurable signals for improving pretraining and evaluation.

3 Related Work

Multilingual pretrained models. Multilingual encoders such as mBERT and XLM-R have demonstrated strong performance on cross-lingual transfer and multilingual benchmarks [3, 2]. These models are commonly evaluated on curated datasets where inputs are monolingual and well-formed.

Robustness to noise and distribution shift. A growing body of work studies robustness of NLP models under input perturbations, including synthetic noise, lexical substitutions, and domain shift. Prior work has shown that models can be sensitive to perturbations that preserve meaning but alter surface form [1, 4]. Such sensitivity is often amplified in informal user-generated content.

Code-switching in NLP. Code-switching is a well-studied sociolinguistic phenomenon and has motivated datasets and tasks in multilingual NLP. However, robustness evaluation under controlled code-switching remains underexplored compared to monolingual corruption benchmarks. Existing evaluations often conflate multiple factors: changes in topic distribution, label noise from weak supervision, or uncontrolled mixing patterns.

Paired evaluation as a robustness tool. SwitchMixBench uses paired evaluation: each clean example is mapped to a perturbed counterpart designed to preserve semantics and labels. This design reduces confounds and allows direct estimation of robustness gaps. Paired evaluation is particularly valuable when the goal is to isolate the impact of a specific perturbation family (here: code-switching and lightweight informal noise) while keeping task semantics constant.

Mechanistic analysis in robustness studies. Many benchmark papers report performance deltas but do not analyze why failures occur. SwitchMixBench explicitly includes diagnostics at the tokenization level and latent representation level. This enables a more mechanistic view of robustness, distinguishing between: (i) input-space shifts induced by tokenization instability, and (ii) changes (or absence of changes) in global sentence representations, which can point to downstream decision instability.

Positioning. SwitchMixBench complements existing multilingual evaluation suites by focusing on realistic multilingual mixing phenomena and by providing analysis modules that can be used to develop robustness-aware pretraining and evaluation strategies.

4 Methodology

SwitchMixBench is built around the idea of *controlled, label-preserving perturbations* applied to multilingual task inputs. The methodology is designed to isolate robustness under code-switching without introducing label noise.

4.1 Paired evaluation framework

Let $(x, y) \sim \mathcal{D}$ denote an input-label pair drawn from a clean multilingual dataset $\mathcal{D}$. We define a perturbation operator $P(\cdot)$ that produces a perturbed input $\tilde{x}$:

$$\tilde{x} = P(x).$$

The perturbation is constructed so that task semantics are preserved and the ground-truth label remains unchanged:

$$y(\tilde{x}) = y(x).$$

A model f_θ is evaluated on both clean and perturbed inputs:

$$\hat{y} = f_\theta(x), \quad \tilde{y} = f_\theta(\tilde{x}).$$

4.2 Robustness gap definition

We define the robustness gap Δ as the expected change in loss (or metric) under perturbation:

$$\Delta = E_{(x,y)\sim\mathcal{D}} \left[\ell(f_\theta(x), y) - \ell(f_\theta(\tilde{x}), y)\right],$$

where ℓ is the task-specific evaluation loss or metric (e.g., accuracy for NLI). This definition is label-preserving by construction and directly quantifies distribution shift induced by perturbation.

4.3 Perturbation design goals

SwitchMixBench perturbations are designed with three constraints:

(i) Semantic preservation. The perturbed input should preserve the original meaning sufficiently so that the task label is unchanged. For NLI, this means that premise/hypothesis relations remain the same. For QA, the question intent and answer span remain aligned.

(ii) Controlled intensity. Perturbations are parameterized to enable systematic sweeps of intensity (e.g., fraction of tokens switched, noise rate). This supports ablation and scaling studies.

(iii) Realism. Perturbations aim to approximate real multilingual usage: lexical mixing, missing diacritics, informal abbreviations, and light orthographic variation.

Perturbation Type	Operator	Example
Code-switching	Lexical substitution	médecin → doctor
Diacritic removal	Character edit	élève → eleve
Insertion	Character edit	famille → fammille
Deletion	Character edit	dîner → diner
Abbreviation	Informal variant	parce que → pcq

Table 1: SwitchMix perturbation operators applied to construct paired inputs.

4.4 SwitchMix perturbation family

We describe the SwitchMix perturbation family as a composition of multilingual mixing and lightweight noise:

$$P(x) = N(S(x)),$$

where $S(\cdot)$ performs code-switching substitutions and $N(\cdot)$ injects informal noise.

4.4.1 Code-switching substitutions

Given a source language L_s and target language L_t (e.g., French $\rightarrow$ English), we identify substitutable lexical units (words or phrases) and replace them with target-language alternatives. Substitutions are applied stochastically, controlled by a mixing ratio $r \in [0, 1]$ that determines the expected fraction of tokens replaced.

Lexicon and mapping. SwitchMixBench supports dictionary-based or rule-based mappings for common lexical items, with extensible hooks for learned mapping models. The current implementation focuses on a French–English setting, but the pipeline is language-pair agnostic in structure.

4.4.2 Informal noise

We consider lightweight perturbations commonly observed in user-generated content:

- missing diacritics (e.g., *"é"* $\rightarrow$ *"e"*),
- character-level edits (single deletion/insertion),
- informal abbreviations and casing variation,
- punctuation simplification.

Noise is controlled by a rate η that determines the probability of applying a given noise operator to a token or character.

4.5 Dataset construction and formats

SwitchMixBench generates paired datasets and stores them in JSONL format. Each record contains:

- `task`: task identifier (`nli`, `qa`),
- `split`: dataset split (`train`, `test`),
- `variant`: `clean` or `perturbed`,
- `input`: model input string or structured prompt,
- `label`: gold label or target,
- `metadata`: perturbation parameters (mixing ratio, noise rate, operators used).

Why JSONL? JSONL supports streaming, large-scale generation, reproducible splits, and easy integration with analysis scripts. It also enables exact tracking of perturbation metadata for ablations and error analysis.

4.6 Mechanistic analysis protocol

To move beyond descriptive performance drops, SwitchMixBench includes analysis modules that quantify:

4.6.1 Tokenizer-Level Distribution Shift

We analyze the effect of code-switching perturbations on subword segmentation by comparing the tokenization of clean and perturbed inputs. Specifically, we measure changes in token count, fragmentation statistics, and token-per-character ratios, as well as the Jensen–Shannon divergence between token distributions induced by perturbation.

These metrics provide an input-space view of distribution shift introduced prior to contextual encoding. Even when semantic equivalence is preserved at the sentence level, multilingual lexical substitutions may alter token boundaries and increase the number of subword units produced by the tokenizer. Such changes effectively inflate the sequence length processed by downstream attention mechanisms.

Because Transformer models operate on tokenized sequences rather than raw text, this tokenizer-level shift constitutes a structural perturbation of the model input that may influence subsequent contextual processing despite minimal semantic change. This provides an input-space view of distribution shift induced before contextual encoding.

4.6.2 Latent Representation Stability

To determine whether code-switching induces instability in the encoder representation space, we extract contextual sentence embeddings at each encoder layer and compute cosine similarity between clean and perturbed inputs. This provides a layer-wise estimate of representation drift across the model.

By comparing embeddings derived from semantically equivalent inputs, we are able to assess whether multilingual perturbations propagate into large-scale changes in the latent space. This analysis enables us to distinguish between robustness failures caused by representational collapse and those arising from downstream decision mechanisms.

4.6.3 Efficiency implications

We measure sequence length deltas and estimate activation memory changes. We also measure inference latency under controlled settings to determine whether robustness failures are accompanied by computational overhead.

4.7 Reproducibility and configuration

All dataset generation and analyses are driven by YAML configuration files, enabling:

- reproducible dataset builds (size, language pair, perturbation intensity),
- standardized evaluation settings (models, max length, batch size),
- consistent output tables for paper writing and comparison.

Output artifacts. The pipeline exports tables (CSV) that can be directly referenced in the paper, including tokenizer statistics, representation shift summaries, robustness metrics, and efficiency measures.

5 Experimental Setup

5.1 Tasks

We evaluate multilingual robustness on two task families: Natural Language Inference (NLI) and extractive Question Answering (QA). For NLI, the model is required to predict the semantic relationship between a premise and a hypothesis. For QA, the model predicts an answer span given a question and a supporting context.

These tasks were selected because they rely on lexical cues and compositional reasoning, making them sensitive to perturbations that alter token boundaries while preserving semantic content.

5.2 Dataset Construction

Paired clean and perturbed inputs are constructed from multilingual task datasets using the SwitchMix perturbation pipeline. Each clean example x is transformed into a perturbed counterpart $\tilde{x}$ using a composition of code-switching substitutions and lightweight informal noise operators.

Perturbations are applied stochastically using a mixing ratio r, which determines the expected proportion of tokens replaced by cross-lingual alternatives. Informal noise is controlled by a parameter η that governs the probability of applying character-level edits such as diacritic removal, insertion, or deletion.

All generated pairs are stored in JSONL format with associated metadata describing perturbation parameters. This allows reproducible evaluation and facilitates ablation studies.

5.3 Model Evaluation

We evaluate multilingual Transformer models under identical inference settings on both clean and perturbed inputs. For encoder-based models such as XLM-RoBERTa, sentence-level representations are extracted using CLS pooling. For sequence-to-sequence models such as FLAN-T5, task predictions are obtained through prompt-based generation.

Evaluation is performed independently on clean and perturbed datasets, and task-level metrics are computed for both variants. The robustness gap is defined as the difference in task accuracy between clean and perturbed inputs.

5.4 Representation Analysis

To quantify latent representation stability, we extract contextual sentence embeddings at each encoder layer of XLM-RoBERTa for paired clean and perturbed inputs. Cosine similarity is computed between representations $\phi_\ell(x)$ and $\phi_\ell(\tilde{x})$ for each layer ℓ.

This provides a layer-wise estimate of representation drift induced by multilingual perturbations and enables assessment of whether performance degradation is accompanied by instability in the encoder representation space.

5.5 Tokenizer Analysis

We analyze the effect of code-switching on subword segmentation by comparing tokenized representations of clean and perturbed inputs. Metrics include average token count, tokens-per-character ratio, and Jensen–Shannon divergence between token distributions.

This analysis captures distributional shifts introduced prior to contextual encoding and provides an input-level view of perturbation effects.

5.6 Efficiency Metrics

We evaluate the computational impact of perturbations by measuring changes in sequence length and estimating activation memory usage. Activation memory is approximated as a function of hidden size and effective sequence length processed by the encoder.

Inference latency is measured in milliseconds per example under CPU-bound evaluation settings, averaged over paired clean and perturbed inputs.

5.7 Implementation Details

All experiments are conducted using configuration-driven scripts provided by the SwitchMixBench toolkit. Dataset construction, evaluation, and analysis are executed using YAML configuration files to ensure reproducibility.

Analysis outputs are exported as CSV files and aggregated into summary tables used for reporting task performance, tokenizer statistics, representation similarity, and efficiency measures.

Parameter	Value
Language Pair	FR–EN
Task	NLI
Model (Encoder)	XLM-RoBERTa
Model (Seq2Seq)	FLAN-T5-Small
Mixing Ratio r	0.2
Noise Rate η	0.05
Max Sequence Length	256
Hidden Size	768

Table 2: Evaluation configuration used for SwitchMixBench experiments.

6 Results

6.1 Tokenizer-Level Distribution Shift

Across 500 paired NLI examples tokenized with XLM-RoBERTa, clean inputs yield an average of 55.04 tokens, while perturbed counterparts produce 55.73 tokens on average. This corresponds to a mean increase of 0.69 tokens per sequence and an average length inflation factor of 1.012.

We further quantify the distributional shift induced by perturbation using the Jensen–Shannon divergence between token distributions of clean and perturbed inputs, obtaining an average divergence of 0.0056 bits.

Because multilingual Transformer models operate on subword token sequences, such perturbations introduce input-level discrepancies prior to contextual encoding:

$$T(x) \neq T(\tilde{x})$$

Metric	Clean	SwitchMix
Avg. Tokens	55.04	55.73
Tokens/Char	0.3270	0.3272
JS Divergence	0.0056 bits	

Table 3: Tokenizer statistics under clean and code-switched inputs.

6.2 Latent Representation Stability

We compute cosine similarity between contextual sentence representations extracted from each encoder layer of XLM-RoBERTa for clean–perturbed input pairs.

Across 200 examples, mean cosine similarity exceeds 0.999999 across all layers, with average drift on the order of 10^{-7}.

This indicates that global sentence-level representations remain stable under code-switching perturbations:

$$\phi_\ell(x) \approx \phi_\ell(\tilde{x})$$

suggesting that performance degradation cannot be attributed to large-scale representational drift.

Layer	Mean Cosine Similarity	Mean Drift
0	0.99999976	2.38×10^{-7}
1	0.99999988	1.21×10^{-7}
2	0.99999979	2.15×10^{-7}
3	0.99999991	8.25×10^{-8}

Table 4: Layer-wise representation similarity between clean and perturbed inputs.

6.3 Robustness Gap under Controlled Code-Switching

On the NLI test set, the FLAN-T5-Small model achieves an accuracy of 0.191 on clean inputs, which drops to 0.030 under SwitchMix perturbation. This corresponds to an absolute robustness gap of 0.161 and a relative degradation of approximately 84%.

This degradation occurs despite the preservation of sentence-level representations, indicating that robustness failures arise downstream of global encoding.

Condition	Accuracy
Clean Input	0.191
SwitchMix Input	0.030
Robustness Drop	0.161

Table 5: Task accuracy on clean and SwitchMix-perturbed inputs for NLI.

6.4 Efficiency Implications

Code-switching perturbations induce a measurable increase in effective sequence length, which in turn leads to a higher number of contextualized token embeddings processed by the encoder.

This increase is reflected in the estimated activation memory footprint, which grows under perturbation despite preserving semantic content.

Interestingly, this internal computational overhead does not necessarily translate into a significant increase in inference latency under CPU-bound evaluation settings. This suggests that robustness failures may arise without obvious runtime indicators.

Metric	Clean	SwitchMix
Seq. Length	55.14	55.82
Latency (ms)	76.73	75.07
Activation Bytes	168960	172032

Table 6: Efficiency metrics under clean and perturbed inputs.

6.5 Qualitative Analysis

To illustrate the effect of code-switching perturbations on model predictions, we present representative examples from the NLI evaluation set.

Example 1 Clean Premise: "Le médecin examine le patient dans la salle." Clean Hypothesis: "Le docteur regarde le malade."

Perturbed Premise: "Le doctor examine le patient dans la salle." Perturbed Hypothesis: "Le docteur regarde le malade."

Although semantic equivalence is preserved, the multilingual substitution of the lexical item "médecin" with "doctor" results in an incorrect prediction by the evaluated model.

Example 2 Clean Premise: "Elle prépare le dîner pour sa famille." Clean Hypothesis: "Elle cuisine un repas."

Perturbed Premise: "Elle prépare le dinner pour sa famille." Perturbed Hypothesis: "Elle cuisine un repas."

Again, semantic equivalence is preserved, yet the perturbed input leads to misclassification.

7 Limitations

SwitchMixBench is currently limited by the use of rule-based perturbation methods for generating code-switched inputs. While this design enables controlled, label-preserving transformations that are suitable for robustness evaluation, it may not fully capture the diversity and contextual variability of naturally occurring multilingual mixing patterns observed in real-world communication.

In particular, spontaneous code-switching often depends on discourse-level factors, speaker intent, sociolinguistic context, and lexical availability, which are difficult to approximate using deterministic substitution rules. As a result, the perturbations generated in the current benchmark should be interpreted as controlled approximations of multilingual distribution shift rather than fully realistic simulations of user-generated mixed-language input.

Additionally, the present benchmark focuses on a single language pair (French–English) and relies on a limited number of seed examples for dataset generation. While the paired evaluation protocol enables systematic robustness measurement, the resulting datasets remain constrained in linguistic diversity and perturbation coverage.

Finally, the current implementation evaluates robustness primarily through performance metrics and lightweight mechanistic analysis. Further work is required to incorporate learned perturbation models, expand language coverage, and conduct fine-grained attention-level or token-level attribution studies to better understand the internal failure modes of multilingual models under code-switching.

Future work will therefore extend SwitchMixBench along three axes: (i) inclusion of typologically diverse language pairs, (ii) integration of learned perturbation generators trained on naturally code-switched corpora, and (iii) deeper analysis of attention dynamics under multilingual perturbations.

8 Implications for Multilingual Pretraining

The robustness failures observed in this study raise important questions regarding the interaction between multilingual pretraining objectives and subword tokenization schemes.

Multilingual language models are typically trained using shared subword vocabularies constructed over heterogeneous multilingual corpora. This results in uneven token coverage across languages and increased fragmentation for low-frequency lexical units.

Under code-switching, lexical substitutions across languages may therefore produce subword segmentations that are underrepresented in pretraining data. While contextual encoders appear capable of integrating such perturbations into stable global representations, downstream decision layers may remain sensitive to token-level distributional shifts.

This suggests that robustness under multilingual perturbation may require pretraining strategies that explicitly account for cross-lingual lexical mixing, such as perturbation-aware objectives or adaptive tokenization mechanisms.

9 Implications Beyond Translation: Multilingual Robustness in Deployment Domains

Multilingual robustness is often discussed in the context of translation accuracy or cross-lingual transfer benchmarks. However, modern language models are increasingly deployed as components of decision-support systems rather than as standalone translation tools. In these deployment settings, model outputs may influence recommendations, classifications, or automated guidance provided directly to users.

Crucially, real-world multilingual inputs frequently include informal orthographic variation and code-switching that are not reflected in benchmark-style evaluation data. This creates a gap between benchmark performance and deployment reliability. Our findings suggest that code-switching can induce measurable tokenizer-level perturbations while leaving global sentence-level contextual representations largely unchanged, yet still produce substantial degradation in downstream task performance.

This raises the possibility that multilingual language models may process mixed-language inputs in a way that preserves semantic representations internally, but still produce unstable task-level predictions. In application domains where model outputs influence user decisions or access to services, such prediction instability may have broader implications beyond translation quality.

9.1 Healthcare and Patient-Facing Systems

Healthcare applications increasingly integrate language models into patient-facing interfaces such as symptom checkers, appointment triage systems, medication guidance assistants, and discharge instruction tools. In multilingual communities, patients frequently communicate using mixed-language input that combines official administrative languages with regional or local languages.

For example, a patient may describe symptoms using a combination of French and a local language or borrow medical terminology from another language where lexical equivalents are unavailable. While such

inputs may preserve semantic intent, they can introduce tokenization-level perturbations that alter subword segmentation and lexical alignment.

Our results suggest that these perturbations may not significantly disrupt global contextual representations, yet still affect task-level decision mechanisms. In healthcare applications, this may influence automated urgency classification, symptom interpretation, or recommendation routing. Importantly, such prediction instability may occur without observable runtime indicators such as increased latency or decoding difficulty, making reliability failures difficult to detect through system monitoring alone.

Evaluating multilingual robustness under deployment-relevant perturbations therefore becomes relevant for ensuring equitable and reliable access to digital health services in linguistically diverse populations.

9.2 Education Technology and Learning Assistance

Language models are widely deployed in educational technology platforms for automated tutoring, question answering, feedback generation, and writing support. Students in multilingual environments frequently switch between languages when describing problems or requesting clarification, particularly in informal learning contexts.

Code-switching may disrupt lexical alignment between instructional prompts and task-specific decision boundaries without altering the underlying semantic content of the query. As a result, tutoring systems may misinterpret the student's intent or provide incomplete explanations despite correctly encoding the global meaning of the input.

In open-ended tasks such as problem-solving assistance or conceptual explanations, such prediction instability may lead to degraded learning support or inconsistent feedback across linguistically diverse users. Robustness evaluation under multilingual perturbation is therefore relevant for ensuring consistent educational assistance in settings where multilingual expression is common.

9.3 Public Services and Administrative Access

Public-facing digital services increasingly rely on automated language processing for information retrieval, eligibility guidance, and application assistance. In multilingual administrative contexts, users may submit queries that combine multiple languages or include informal orthographic variants, particularly when navigating unfamiliar bureaucratic terminology.

Prediction errors in these settings may influence access to information about legal procedures, public benefits, or social services. Since such systems often operate as the first point of contact for service access, robustness under realistic multilingual input conditions becomes important for ensuring fairness and usability across populations with varying linguistic backgrounds.

Moreover, because performance degradation under code-switching may not manifest through runtime indicators such as latency or output length, prediction instability may remain undetected during deployment.

9.4 Customer Support and Financial Advisory Systems

Customer support and financial advisory platforms increasingly deploy conversational agents for account management, billing support, and transaction guidance. Multilingual users may submit mixed-language queries related to account access or dispute resolution, often including domain-specific terminology borrowed from multiple languages.

Prediction instability in these contexts could affect automated routing decisions, fraud detection workflows, or eligibility assessments for financial services. As in healthcare and public services, robustness failures may not be observable through system-level performance metrics, complicating monitoring and quality assurance processes.

9.5 From Translation Quality to Decision Reliability

Across these domains, the central concern extends beyond translation fidelity to task-level decision reliability. The key question becomes whether multilingual language models maintain stable predictions under meaning-preserving perturbations that reflect realistic deployment conditions.

Our results indicate that code-switching can introduce measurable tokenizer-level distribution shifts without destabilizing global sentence-level representations, while still producing substantial performance degradation on downstream tasks. This suggests that multilingual robustness should be evaluated not only as a property of language understanding, but as a reliability constraint on decision-support systems operating in linguistically diverse environments.

SwitchMixBench therefore provides a framework for assessing robustness under multilingual perturbation in contexts where prediction errors may influence user outcomes, access to services, or automated recommendations beyond the scope of translation tasks.

10 Discussion

10.1 Summary of Empirical Findings

The SwitchMixBench evaluation reveals a substantial robustness gap under controlled code-switching perturbations. On the NLI task, we observe a significant drop in task accuracy when moving from clean inputs to their SwitchMix counterparts, despite the preservation of sentence-level semantics by construction.

Beyond performance degradation, the benchmark analysis provides insight into the internal behavior of multilingual models under perturbation. Specifically, we observe that code-switching induces measurable tokenizer-level changes while leaving global sentence-level contextual representations largely invariant across encoder layers.

Taken together, these findings suggest that robustness failures cannot be attributed solely to large-scale instability in the latent embedding space.

10.2 Tokenizer-Level Effects

Code-switching systematically alters subword segmentation, producing small but consistent increases in sequence length and shifts in token distributions. While these perturbations preserve the semantic content of the original input, they introduce localized discrepancies in token identity and position prior to contextual encoding.

Because multilingual Transformers operate on tokenized sequences rather than raw text, such discrepancies may influence how downstream attention mechanisms process task-relevant information, even when overall meaning is retained.

10.3 Representation Stability

Layer-wise cosine similarity between contextual sentence representations of clean and perturbed inputs remains extremely high across encoder layers. This indicates that multilingual encoders are able to integrate cross-lingual lexical substitutions into globally consistent latent representations.

Formally, for semantically equivalent inputs x and $\tilde{x}$, we observe:

$$\phi_\ell(x) \approx \phi_\ell(\tilde{x})$$

for all encoder layers ℓ, where ϕ_ℓ denotes the contextual representation at layer ℓ.

This stability suggests that performance degradation under code-switching does not stem from representational collapse in the encoder.

10.4 Downstream Decision Instability

Despite stable contextual embeddings, downstream task accuracy drops substantially. This indicates that robustness failures may arise downstream of global encoding, potentially through localized perturbations that affect token-level feature extraction or classification boundaries.

In particular, token substitutions that disrupt learned lexical alignments or positional attention patterns may influence decision layers without inducing large-scale representational drift.

We therefore hypothesize a failure pathway of the form:

$$\text{Tokenizer-Level Shift} \rightarrow \text{Attention Misalignment} \rightarrow \text{Decision Instability}$$

under which multilingual models maintain global semantic coherence while exhibiting brittle task-level behavior.

10.5 From Stable Representations to Unstable Decisions

The coexistence of near-invariant sentence-level contextual representations and substantial task-level performance degradation suggests that multilingual robustness failures do not arise from global semantic collapse in the encoder latent space.

Instead, code-switching appears to introduce localized token-level perturbations that alter lexical alignment patterns relied upon by downstream task heads or decoding mechanisms. Because Transformer-based decision layers operate on tokenized contextual embeddings, small changes in subword segmentation or token identity may influence attention allocation or classification boundaries even when the pooled sentence representation remains stable.

In other words, semantically equivalent inputs x and $\tilde{x}$ may occupy similar regions of the encoder representation space while differing in token-level feature configurations that are critical for downstream decision rules. This implies that prediction instability can emerge independently of large-scale embedding drift.

This distinction is particularly relevant in deployment settings, where internal representation similarity does not necessarily guarantee consistent task-level behavior. Multilingual perturbations that preserve meaning may therefore degrade prediction reliability without affecting global semantic encoding, complicating robustness monitoring in practical applications.

10.6 Efficiency Considerations

SwitchMix perturbations slightly increase effective sequence length and estimated activation memory footprint. However, this does not necessarily translate to a significant latency penalty in CPU-based inference settings.

This finding is particularly relevant in deployment scenarios: robustness failures may occur without obvious computational warning signs such as increased response time.

10.7 Implications for Multilingual Pretraining

The observed robustness gap raises questions regarding the interaction between multilingual pretraining data and shared subword vocabularies. Uneven token coverage across languages may lead to increased fragmentation for low-frequency lexical units under code-switching.

While contextual encoders appear capable of integrating such perturbations into stable global representations, downstream decision layers may remain sensitive to token-level distributional shifts.

This suggests that improving robustness under multilingual perturbation may require pretraining strategies that explicitly account for cross-lingual lexical mixing or perturbation-aware objectives.

11 Mechanistic Interpretation

The experimental findings presented in this work reveal a decoupling between tokenizer-level perturbations and latent representation stability in multilingual Transformer models under controlled code-switching.

While code-switching introduces systematic changes in subword segmentation that increase effective sequence length and alter token distributions, we observe near-invariant contextual representations across encoder layers. In particular, cosine similarity between clean and perturbed sentence embeddings consistently exceeds 0.999999, indicating that global semantic representations remain stable under perturbation.

Formally, let $T(\cdot)$ denote the tokenizer and $\phi_\ell(\cdot)$ the contextual encoder representation at layer ℓ. For semantically equivalent inputs x and $\tilde{x}$, we observe:

$$T(x) \neq T(\tilde{x}), \quad \text{but} \quad \phi_\ell(x) \approx \phi_\ell(\tilde{x})$$

This suggests that multilingual encoders successfully integrate cross-lingual lexical substitutions into globally coherent latent representations.

Despite this stability, downstream task performance degrades substantially, with observed accuracy drops of up to 84% on NLI under controlled perturbation. This indicates that robustness failures arise downstream of contextual encoding rather than from representational collapse.

We hypothesize that token-level perturbations induced by code-switching disrupt learned lexical alignments or positional attention patterns in task-specific decision layers. Such localized perturbations may affect classification outcomes without inducing large-scale embedding drift.

Taken together, these findings support a failure pathway of the form:

$$\text{Token Inflation} \rightarrow \text{Attention Dispersion} \rightarrow \text{Decision Instability}$$

in which multilingual models maintain global semantic coherence yet exhibit brittle task-level behavior under localized multilingual perturbations.

12 Future Work

SwitchMixBench is designed as a diagnostic benchmark and toolkit, and several research directions naturally follow from the current results. We outline future work along four axes: language coverage, perturbation realism, deeper mechanistic analysis, and robustness interventions.

12.1 Extending Language Coverage and Typological Diversity

A first priority is to expand beyond the current French–English setting. Robustness under code-switching is likely to vary substantially across language pairs due to differences in morphology, script, word segmentation, and shared vocabulary coverage in multilingual tokenizers. Future versions of SwitchMixBench will therefore incorporate typologically diverse language pairs, including settings with: (i) higher morphological complexity, where subword fragmentation may increase, (ii) script changes (e.g., Latin vs non-Latin scripts), which may induce stronger tokenization discontinuities, and (iii) low-resource languages that are underrepresented in pretraining corpora.

In addition, we plan to include West African language settings that reflect real deployment needs (e.g., French mixed with regional languages), subject to the availability of suitable lexical resources and evaluation data. This expansion is critical to ensure that robustness evaluation reflects multilingual realities beyond high-resource language pairs.

12.2 Learned Perturbation Models and Naturalistic Code-Switching

The current benchmark relies on rule-based perturbation operators, which offer strong control and label preservation but may not capture the diversity of naturally occurring code-switching. A second direction is therefore to incorporate learned perturbation models trained on real code-switched corpora (when available). Such models could generate more realistic mixing patterns conditioned on context, syntactic structure, and discourse cues.

A key challenge will be maintaining label preservation when perturbations become more naturalistic. Future work will investigate constrained generation strategies that preserve meaning while introducing realistic code-switch structure, as well as human-in-the-loop validation protocols for quality control on small evaluation subsets.

12.3 Mechanistic Analysis Beyond Pooled Representations

Our representation analysis suggests that pooled sentence embeddings remain nearly invariant under perturbation, despite large drops in task performance. This motivates deeper mechanistic analysis focused on token-level pathways. Future work will examine: (i) attention maps and attention entropy under clean vs perturbed inputs, (ii) attribution methods to identify which tokens and subwords drive decisions, and (iii)

layer-wise sensitivity analysis that probes where decision-relevant information diverges even when pooled embeddings remain stable.

Such analyses may help distinguish between competing explanations of failure, including token boundary sensitivity, lexical shortcut reliance, and prompt-format brittleness for instruction-tuned models.

12.4 Robustness Interventions and Benchmark-Driven Training

Finally, SwitchMixBench can serve as a testbed for robustness interventions. Future work will evaluate training and inference-time strategies such as: (i) perturbation-aware data augmentation that includes controlled code-switching during fine-tuning, (ii) tokenizer adaptations (e.g., vocabulary augmentation or language-aware tokenization) to reduce fragmentation under mixing, and (iii) decision regularization objectives that encourage prediction stability under meaning-preserving perturbations.

Importantly, these interventions can be evaluated under the same paired, label-preserving protocol, enabling direct measurement of whether improvements reflect genuine robustness gains rather than dataset artifacts.

13 Conclusion

This paper introduced SwitchMixBench, an open-source benchmark and evaluation toolkit for measuring multilingual model robustness under controlled code-switching and informal noise perturbations. SwitchMixBench constructs paired clean and perturbed inputs that preserve task labels, enabling direct estimation of robustness gaps under a targeted, label-preserving distribution shift.

Empirically, we observe a substantial degradation in NLI accuracy when moving from clean inputs to SwitchMix-perturbed inputs. To better understand why this degradation occurs, we complemented standard performance reporting with mechanistic analyses. We showed that code-switching induces measurable tokenizer-level changes, including sequence length inflation and a shift in token distributions, while layer-wise sentence-level representations remain nearly invariant under perturbation. This decoupling suggests that robustness failures arise downstream of global encoding, likely through localized token-level effects that perturb decision mechanisms without inducing large-scale embedding drift.

Taken together, these findings highlight a key limitation of current multilingual evaluation practices: strong benchmark performance on clean, monolingual data may not translate to reliability under realistic multilingual usage. SwitchMixBench provides a controlled, reproducible framework for diagnosing these failures and for evaluating future robustness interventions. We hope this benchmark will support more realistic multilingual evaluation and contribute to the development of multilingual models that are robust in the settings where they are most needed.

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